

1. a) **if** (Grade >= 90)

System.out.println("Great Job!");

- b) if (number < 20 | number > 50)

System.out.println("Error");

- c) **if** (y < 100)

y = y + 2;

2. **if** (num1 < num2)

System.out.println("Number one is bigger than number 2");

else (num2 < num1)

System.out.println("Number two is bigger than number one");

else (num1 = num2)

System.out.println("Number two is bigger equal to number 1");

3. a) even , odd

b) switch (num % 2) {

Case 0:

System.out.println(" even number");

break;

Case 1:

System.out.println(" odd number");

Break;

Case -1:

System.out.println(" odd number");

Break;

}

4. a) Int randomNum = (int) (Math.random() *49) +1

b) **int** randomNum = (**int**) (Math.random() *80) +20;

c) **double** randomNum = (**double**) (Math.random() *10) +10;

5. In the third option for the if statement it is else if when it should just be else.

6. a) true

b)false

c)false

d)false

e>true

f>true

g>false

8. a) true

b) false; a regular **if** else statement would stop if the first **if** statement is true vs a nested **if** statement would continue to the next statement if the first one is true.

c) false; the expression of a switch statement doesn't have to even be a number, it could be a word like be defined as string instead of int or double.

d) false; I would say that they aren't random but I don't exactly know if they're pseudorandom as I don't quite understand what that means. Numbers generated by a computer program are defined or created by the programmers rules not based on random factors.

e) false; the double cast wouldn't even generate a integer but a number with a decimal place value.

f) true

g) true

h) false; ! takes higher precedent over &&.

i) true

j) true